RESUME

Youngdu Min · Unity Game Client Programmer

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Write-up · How I moved 75%+ of an education app's content onto one core architecture

Unity client programmer with 3 years of experience. On "ToctocMath," used by about 6,900 schools, I designed the core/base scripts the team built its content on during major updates and cut build size by ~70%. I also shipped a Steam title as lead programmer of a 7-person team and now build multiplayer games on a 5-person indie team.

Experience

Cemware · Unity Developer · Aug 2022 – Jul 2025 (3 years, incl. alternative military service)

ToctocMath · KERIS digital-textbook-linked math education app (incl. mini-games), live service · 8 developers (up to 12 with interns), lead programmer · iOS / Android / WebGL

- Designed the core/base scripts for the content flow (question → interaction → grading → saving) in the major updates; 75%+ of content, all but standalone pieces, was built or migrated onto them. Legacy content migrated by inheriting the base and deleting duplicate code; shared interactions (drag-and-drop, graphs) became modules
- Took the lead on the team's game features: built 8+ casual educational mini-games and reworked most existing ones for curriculum changes and live issues
- Contributed to 167% YoY download growth and the 1st-place 2025 Korea Quality Satisfaction Award (Elementary Math) by improving core systems around teachers' needs and optimizing app size and performance
- Cut build size from over 1 GB to under 350 MB (~70%) across all platforms by replacing assets with reusable sliced UI sprites, removing unused assets, and building an editor tool for batch texture compression
- Optimized WebGL memory (nearly 2 GB per tab) that froze low-end school Chromebooks with several tabs open, using Addressables among other fixes (no further reports); redesigned a drawing feature that stacked a full-screen copy per stroke into a cache-and-update structure, fixing mobile crashes
- Integrated LMS embedding and AI backend communication (learning analytics, problem recommendation)
- Led Git workflow and releases (store review, WebGL CDN deployment); turned a plain-text build procedure into a Notion checklist teammates could follow alone, and automated steps with batch scripts and a build-check editor tool

Shipped Games

Hollow: Cook Off · Side project · 2D action side-scroller (Corgi Engine; 12 stages + 5 boss stages) · Lead programmer (overall programming), team of 7 · 2024 · Steam · Trailer · Feature demo · Code

- Extended Corgi Engine's AI Brain with boss-phase management and State/Decision tracking & recovery
- Derived many custom AI Actions and Decisions to build AI for 9 enemy types and 5 multi-phase bosses (randomized attack-pattern groups per phase, a healer AI that repairs allies' armor, etc.)
- Implemented weapons and abilities the engine did not support: bullet parry with slow motion, grappling gun, boomerang
- Built a JSON-based dialogue system, Korean/English localization, a save system and Timeline cutscenes
- Owned the full Steam release pipeline: build, review and release

Karma Defence · Defense · Team of 5, sole programmer · 2020 – 2021 · Google Play (service ended) · Code

- Built all programming, including multi-type unit object pooling (pool doubles when exhausted), stage quests with star rewards, upgrades, and a conditional inspector tool for designers

Others

- **Keyboard Invader** (team of 3, lead programmer, 2021): 2D shooter where each key fires a unique bullet; built a 2-phase boss AI and the shop; Excellence Award (2nd place), SBCF 2021 Game Category · Video · Play
- **Gun Slash** (solo, itch.io 2021, Corgi Engine): Local multiplayer shooter for up to 4 players with a lobby and custom input system supporting 2 keyboards + 4 gamepads · Video · Play

- **Gravity Miner** (university team of 7, main programmer, 2019 – 2020): Core gameplay and physics for a stomp-combat platformer · Video · Play
- Game jams: Beaver Jam 2023 Option Adventure (lead programmer), GMTK 2022 Dice Knight and 2020 Dungeon Keyboarder

Technical Projects

Indie Game Team · Programmer on a team of 5 (2 design · 2 programming · 1 art) · Unity 6 / URP / FishNet · Mar 2026 – present

- **Data-Driven Monster AI** (Mar – Apr 2026, Video): Modular FSM with inspector-defined transition rules, sight/distance detection, A* movement and non-lethal stabilization; 3 monster types differ only in their rules
- **Asymmetric PvP** (May – Aug 2026, Backrooms-style 1v3 top-down, Video): Server-authoritative escape system, a player-controlled monster's dash-attack FSM, survivor paths revealed only to the monster via TargetRpc, and a seed-synced 2D procedural map (MST + loops, A* corridors, validation)
- **Procedural 3D Map Baker** (Sep 2026, discontinued, built with Claude Code, Video): Bakes the 2D map into modular 3D prefabs with deterministic host/client placement verified by hashes, Union-Find connectivity, an editor tool that batch-generates 73 wrapper prefabs, and a convolution filter on URP Render Graph

Unit System for Unity · Designed solo within a team project · 2025 (released as a module)

- Reusable unit system with interfaces and ScriptableObject presets: class-based FSM, status effects, per-part hit detection, boss-phase control, Animator-free animation controller

Other Work Projects (Cemware)

- **Pump Simulation Software** (client delivery, team of 2): All UI interaction and simulation logic (Observer pattern)
- **Teaching-aid WebGL Service** (client delivery, team of 3, 2025): 10+ interactive teaching aids; JavaScript device detection to load AssetBundles in each device's texture format
- **ImmersionSim** (AR/VR): AR Foundation / VRTK / MRTK2 content with WebGL ↔ JavaScript save/load and UGC

Core Skills

- **Unity**: Unity 6 / URP, UGUI development & optimization, ScriptableObject architecture, Addressables memory management, NavMesh, Timeline, DOTween, UniTask
- **C# & Architecture**: OOP / SOLID; Observer, Strategy, Factory, FSM and Mediator patterns; legacy code refactoring and library development
- **Gameplay**: Corgi Engine AI extension (9 enemy types, 5 bosses); data-driven FSM AI; weapon, skill, dialogue, quest and interaction systems; object pooling; localization (Korean / English)
- **Procedural Generation**: MST, A* and Union-Find map generation pipeline; seed-based deterministic generation
- **Optimization & Multi-platform**: Unity Profiler, build size cut from 1 GB+ to under 350 MB (~70%), WebGL memory optimization (~2 GB per tab), live operation on iOS / Android / WebGL
- **Networking**: FishNet server-authoritative design (SyncVar / RPC / TargetRpc, late-join sync), seed sync with no map-data transfer, JSON / REST API, WebGL ↔ JavaScript interop
- **Release**: Steam, App Store and Google Play build, review and release; WebGL verified on a test server before CDN deployment
- **Tools & Collaboration**: Custom editor tools (CSV export for reports, prefab finder by root/component, batch texture resizing, prefab wrapper generation); led Git workflow; dev manuals; Jira, Trello, Notion, Obsidian, Figma, Adobe XD, Slack
- **AI-assisted Development**: Claude Code with plan-first review and approval, using references I choose

Education · Certifications · Military Service

- Korea National Open University, B.S. in Computer Science (transfer) · Aug 2025 – Aug 2027 (expected)
- Jangan University, Associate Degree in Game Contents · Mar 2019 – Feb 2022
- Engineer Information Processing (2026), ISTQB CTFL (2019) · Military service completed (2022 – 2024)